

Local-Drive Legendre-Chain Transfer Bounds for a Finite-Temperature Quantum Latch

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Abstract—We retain the exact thermal covariance of a flat-band Gaussian reservoir while truncating its vacuum propagation in a Legendre-chain representation. Local system insertions give a factorial continuum-transfer allowance for the second-order response. With inherited parameters $\beta = 5/16$, $T = 16$, and a conservative excitation-sector norm ceiling 64, a 3072-site reference gives an exact rational allowance below 2^{-451} . This removes the coarse midpoint-embedding allowance from this alternative reference. The second-order response itself remains uncomputed; the primary 2^{-16} thermal target, numerical roundoff certificate, and physical realization remain open. The larger chain is a computational reference, not a physical resource advantage.

Index Terms—Quantum control, open quantum systems, reservoir chain mapping, certified truncation.

I. SOURCE CONTRACT AND SCOPE

The source parameters are $\Omega = 64$, $B = 16$, $\kappa = 8$, $T = 16$, $h = g = 1$, and $\lambda^2 = 4\sqrt{2}$. The retained native basis is $(p0, m0, e0, d1, d0, p1, m1, e1)$, with excitation charges $(1, 1, 1, 1, 0, 2, 2, 2)$. The launch is $|p0\rangle$ with reservoir vacuum; an untouched output bank and arbitrary reference factor are carried identically. The local operator $S = I_C \otimes \sigma_R^-$ has norm one. The source Gaussian continuum reduced dynamics and its coherent-displacement representation are assumed to exist, as in the parent reference [2]. This paper does not derive a microscopic device or a permanent irreversible sink.

The new reference retains the exact stationary covariance

$$C_\beta(s) = \frac{\kappa}{2\pi} \int_{-B}^B \frac{e^{-i\delta s}}{e^{\beta(\Omega+\delta)} - 1} d\delta. \quad (1)$$

Write $V_\beta = C_\beta(0)$; positivity gives $|C_\beta(s)| \leq V_\beta$. The inherited rational envelope is

$$T^2 V_\beta < \frac{16384}{31457265}. \quad (2)$$

No independent thermal reset of chain sites, thermal occupation discretization, or divergent continuum $\text{tr } N$ subtraction is introduced.

II. EXACT CHAIN MAPPING

The orthogonal-polynomial chain mapping is established prior art [1]. For the flat measure, normalized basis functions and recurrence coefficients are

$$\phi_n(\delta) = \sqrt{\frac{2n+1}{2B}} P_n(\delta/B), \quad (3)$$

$$\delta\phi_n = b_{n+1}\phi_{n+1} + b_n\phi_{n-1}, \quad (4)$$

$$b_n = \frac{Bn}{\sqrt{4n^2 - 1}}, \quad n \geq 1. \quad (5)$$

MQM development manuscript v0.1. Conditional derivation and exact constructor controls; independent theorem review pending.

Thus the bath one-particle operator J is a zero-diagonal semi-infinite Jacobi matrix with $\|J\| = B$. Flat system coupling becomes a single port at chain site zero, with $G^2 = \kappa B/\pi = 128/\pi$. Define H_L by cutting the link between sites $L-1$ and L , on the same infinite Fock space as H_∞ . The unused tail starts in vacuum and remains decoupled, so propagation is equivalent to the first L sites.

The vacuum Hamiltonian conserves total excitation Q . All ket and bra trajectories in the second-order paired response use only $Q = 0, 1, 2$. On these sectors both Hamiltonians obey

$$\|H\| \leq 2B + h + g + \lambda + 2\sqrt{2}G < 64 = \Lambda. \quad (6)$$

This follows from $\pi > 3$, $G < 7$, $\lambda < 5/2$, and $\sqrt{2} < 3/2$. The cut bath is a compression of the bounded one-particle operator, with the same norm bound. The one-excitation ceiling 24 from the cold reference is not reused here.

III. LOCAL-DRIVE TRANSFER THEOREM

Let $U(t) = e^{-iHt}$, $S_s = U(s)^\dagger S U(s)$, and $\rho_* = |p0, \text{vac}\rangle\langle p0, \text{vac}|$. The source response is

$$\Delta_2^H = - \int_0^T ds \int_0^s du \mathcal{R}_T^H(C_\beta(s-u)[S_s^\dagger, [S_u, \rho_*]] + C_\beta(u-s)[S_s, [S_u^\dagger, \rho_*]]), \quad (7)$$

where $\mathcal{R}_T^H(X) = \text{tr}_{\text{bath}} U(T) X U(T)^\dagger$. The final evolution is essential. Set $\Delta_{2,L} = \Delta_2^{H_L}$ using the same C_β .

Lemma 1. *Every ket word launched from reservoir vacuum, containing at most two paired local $S, S^\dagger$ insertions and fewer than L Hamiltonian factors, agrees between H_∞ and H_L .*

Proof. Only the port creates a bath excitation at site zero. Each bath hopping factor moves it by at most one site, and system insertions do not change its bath position. Reaching the omitted site L therefore needs at least $L+1$ Hamiltonian factors. The conservative degree cutoff L is sufficient. The argument permits two bath photons and bosonic occupation factors. $\square$

Theorem 1. *Under the source contract above,*

$$\frac{1}{2} \|\Delta_2^{H_\infty} - \Delta_{2,L}\|_1 \leq 8T^2 V_\beta \sum_{n=L}^{\infty} \frac{(\Lambda T)^n}{n!}. \quad (8)$$

This is a fixed-launch second-order response bound, not a certificate for the full thermal channel.

Proof. Expand each commutator into four dyads, giving eight covariance-weighted dyads in total. Rewrite them in the

Schrödinger picture. Each ket and bra has nonnegative propagation segments summing to T , with at most two norm-one local insertions. Multinomial grouping of the segment Taylor expansions bounds total degree n by $(\Lambda T)^n/n!$. By the lemma, all degrees below L agree; a ket difference is at most twice the remaining tail. Each exact ket has norm at most one, so a dyad difference is at most four times that tail. Partial trace is contractive in trace norm. The factor $1/2$, eight dyads, and triangle area $T^2/2$ give the coefficient eight in (8). No scalar phase cancellation is substituted for noncommuting operator propagation. $\square$

IV. EXACT RECEIPT AND RESOURCE COST

For $z = \Lambda T = 1024 = 2^{10}$ and $L = 3072 = 3z$, successive positive Taylor terms have ratio at most $z/(L+1) < 1$. Equations (2)–(8) give the explicit rational bound

$$\delta_{\text{chain},2} < \frac{131072}{31457265} \frac{1024^{3072}}{3072! [1 - 1024/3073]} < 2^{-451}. \quad (9)$$

Integer numerator/denominator comparison establishes the last inequality; no floating-point quadrature is used. The frozen acceptance requirement 2^{-128} therefore passes. Exact integer controls on cuts $L = 2, 3, 4$ give 115 word equalities and three nonzero crossing witnesses at degree $L+1$. These finite controls test the support mechanism; the all-size result rests on the proof.

The $Q = 2$ reference dimension is $3 + 4L + L(L+1)/2 = 4,732,419$. One complex binary64 vector occupies 75,718,704 bytes, excluding workspaces. The combined $Q \leq 2$ space contains 4,735,496 states. No full reference propagation was run. $L = 3072$ polynomial-chain sites and $M = 768$ midpoint frequency bins are different representations; their equal role as numerical modes does not make their operators or thermal states identical.

V. PROMOTION GATE

The remaining sufficient target gate is

$$\frac{1}{2} \|\Delta_{2,L}\|_1 + \delta_{\text{chain},2} + \delta_{\text{num}} < 127/2^{25} - 2^{-128}, \quad (10)$$

using the inherited cold allowance and higher-order Gaussian ceiling. Here δ_{num} must include directed propagation, covariance evaluation, quadrature, and matrix assembly errors. None has been certified for a completed full-response evaluation. A failed upper bound is inconclusive; it does not prove cooling necessary. The earlier midpoint allowance remains valid but insufficient for its original reference. The new chain transfer theorem is conditional, pending independent review; the thermal target stays OPEN and physical promotion stays zero.

REFERENCES

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- [2] B. Mayes, *MQM Thermal Kernel, Frequency Bound, Second-Order Specification, and Sparse Transfer*, development releases v0.1, accompanying source anchors and receipts.